Theo Hytopoulos

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Education

M.S. Computer Science, *Western Washington University* 2024 - 2025
B.S. Computer Science, *Western Washington University* 2022 - 2024

Technical Skills

- **Programming Languages**: Python, SQL, Typescript, C/C++
- **Machine Learning Techniques**: SFT, PEFT, MCP, RAG, RL (GRPO), SAEs
- **Frameworks**: PyTorch, Huggingface, LangChain, SkyPilot, Verl, React, Django, Docker

Experience

- **Contract Automation Engineer** *2025*
  Leviathan Games Remote
  - Constructed and published a GitHub syncing wrapper on n8n.io, enabling version control and collaboration for 200k+ developers.
  - Automated API flows with n8n and LLM integration, eliminating backend glue code and vastly reducing complexity.

- **Research Assistant** *2023 - 2025*
  Western Washington University Bellingham, WA
  - Authored 2 publications and presented research as part of a panel discussion at PROFES 2025, sharing key contributions with an audience of academic and industry professionals from across Europe.
  - Fine-tune Llama embeddings for source code vulnerability detection, increasing upon the human baseline precision by 102%.
  - Conducted a case study and training program, leading to a 154% increase in consensus on the root causes of software vulnerabilities.

- **Software Engineer Intern** *2022*
  Leviathan Games Remote
  - Developed and tested C code on a prototype SoC for a consumer device sold at major retail outlets, focusing on performance and security in an embedding Linux environment.
  - Programmed key file management systems including configuration parsing and loading of user data.

Relevant Projects

- **SigLIP Image Emotion Detection using Sparse Autoencoders** *2025*
  torch, verl
  - Develop a novel method for emotion detection by training a sparse autoencoder on multimodal SigLIP embeddings.
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- **Exact Decompilation Using Reinforcement Learning with Levenshtein Distance Metrics** *2025*
  torch, verl
  - Leveraged group relative policy optimization to train a 7B parameter LLM on source code recovery, beating GPT-4.1 by 13.9% in zero-shot accuracy for C++ decompilation.
  - Designed a custom reward function using Levenshtein distance between gold ASM and compiled C++ response.
  - Utilized verl for parallel reinforcement learning and AWS S3 for cloud compilation, choosing spot GPUs for cost-efficient model training.

- **Few-shot Multi-label LoRA Finetuning for Identifying Human Errors** *2024*
  torch, transformers, peft
  - Fine-tuned quantized 7B Llama model for multi-label few-shot software vulnerability detection.
  - Designed a supervised fine-tuning architecture using end-of-sentence token embedding with linear classifier and weighted cross-entropy loss.